**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 122 · Lesson 122 · Selecting a Development Method: Case Study

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What Boolean operator joins invalid cases score<0 and score>100?

Q2. A linked-list node has data and what other essential field?

Q3. In a BST, where is a value less than the current node placed?

Q4. Why can an array provide direct indexed access?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

functional requirement — Behaviour a system must provide.

non-functional requirement — Quality or constraint on system operation.

acceptance criterion — Condition showing requirement has been met.

trade-off — Choice balancing competing advantages and costs.

## New learning

Analyse the brief — Separate stakeholder requirements, constraints, risk and available resources.

Choose a lifecycle — Justify one method against credible alternatives with evidence.

Trace a core algorithm — Write and test one key decision procedure that fits the brief.

## Worked example

A regional bus service needs journey booking with changing fare rules and a fixed launch date. Choose an approach and an algorithm.

1. Model answer — Use iterative development for changing fare rules with documented acceptance milestones for launch. Fare algorithm: collect zones and passenger category, select valid tariff, compute fare, reject invalid input and test boundary zones. — Check: link each claim to the scenario and use precise OCR terminology.

# We Do · Practise with guidance

A school canteen needs a pilot ordering system before full deployment. Recommend a lifecycle and decision routine.

What must we decide?

Identify the inputs, relevant facts and exact output or judgement required.

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

RAD or agile prototypes gather pupil feedback quickly; keep data security and payment assurance as formal checks. Algorithm checks available stock, valid quantity and eligibility before confirming order.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

State two functional and two non-functional requirements for an event-booking service. [6]

### Question 2

Justify a suitable lifecycle for a well-understood regulated payroll update. [6]

### Question 3

Design an algorithm for rejecting bookings when capacity is full and explain test cases and lifecycle fit. [8]

# Exit Ticket · Answer before checking

The development method should be chosen without reference to users, change risk or deadlines.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. OR.

Q2. A link/reference to the next node (for a singly linked list).

Q3. In its left subtree (under the chosen no-duplicate convention).

Q4. Its element locations can be calculated from base address and index.

### You Do pages · Independent questions

Q1. Functions: search/book/cancel etc [3]; non-functional: accessibility, security, response times, reliability etc [3].

Q2. Documented staged/hybrid development with control [2], traceable tests/compliance [2], limitations or alternative [2].

Q3. Capacity check [2], valid request/quantity [2], normal/full/over-limit tests [2], method choice linked to changes/risks [2].

### Exit Ticket · Model answer

False. Those constraints drive the choice.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**